

The probability scale; calculating and estimating probability

From “likely” to a number between 0 and 1 — by counting, and by experiment.
[Cambridge insert]

LESSON 171–172

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 13 PROBABILITY

LESSON CLOCK

20'

22'

20'

18'

The probability scale	20 min
Counting the outcomes	22 min
Probabilities that add to 1	20 min
Estimating from experiment	18 min

LEARNING OBJECTIVES

- Place an event on the probability scale from 0 to 1.
- Calculate a probability when the outcomes are equally likely.
- Use the fact that all the probabilities of one experiment add to 1.
- Estimate a probability from the results of an experiment.

TEXTBOOK

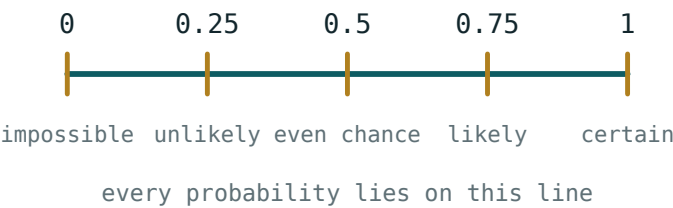
National	pp. 488–492
Cambridge	Stage 7, pp. 132–140
Workbook	Exercise 13.1–13.2

01

Explanation

The probability scale

Every probability is a number from 0 to 1. Nothing is less likely than impossible and nothing more likely than certain, so nothing lies outside.



The probability scale — impossible at 0, certain at 1

EVENT	IN WORDS	PROBABILITY
a die shows 7	impossible	0
a coin shows a head	even chance	$\frac{1}{2}$
a die shows a number under 7	certain	1
a die shows 4	unlikely	$\frac{1}{6}$

A PROBABILITY IS NEVER A NUMBER LIKE 2 OR -0.3

An answer outside 0 to 1 means the counting went wrong — usually the wrong total underneath. Check that number before anything else.

Counting the outcomes

When every outcome is equally likely, a probability is a fraction: how many outcomes count as success, over how many there are altogether.

$$P(event) = \frac{\text{favourable outcomes}}{\text{all outcomes}}$$

EXPERIMENT	EVENT	COUNT	PROBABILITY
one fair die	a 4	1 of 6	$\frac{1}{6}$
one fair die	an even number	3 of 6	$\frac{1}{2}$
one fair die	more than 4	2 of 6	$\frac{1}{3}$
one fair die	a multiple of 3	2 of 6	$\frac{1}{3}$
a bag of 10 counters	red, with 4 red	4 of 10	$\frac{2}{5}$

“EQUALLY LIKELY” IS A REAL CONDITION, NOT A FORMALITY

The fraction rule works for a fair die and for counters drawn without looking. It does not work for “it rains tomorrow”, where the two outcomes are not equally likely at all.

Probabilities that add to 1

A bag holds 4 red, 3 blue, 2 green and 1 yellow counter.

COLOUR	COUNTERS	PROBABILITY	IN LOWEST TERMS
red	4	$\frac{4}{10}$	$\frac{2}{5}$
blue	3	$\frac{3}{10}$	$\frac{3}{10}$
green	2	$\frac{2}{10}$	$\frac{1}{5}$
yellow	1	$\frac{1}{10}$	$\frac{1}{10}$
total	10	$\frac{10}{10}$	1

$$P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{2}{5} = \frac{3}{5}$$

SUBTRACTING FROM ONE SAVES A LOT OF COUNTING

“Not a six” has five favourable outcomes, but $1 - \frac{1}{6} = \frac{5}{6}$ is quicker and safer — and for a large set of outcomes it is the only sensible route.

Estimating from experiment

Some probabilities cannot be counted, because the outcomes are not equally likely. Then the experiment itself is the evidence: the **relative frequency** estimates the probability.

relative frequency = $\frac{\text{times it happened}}{\text{times it was tried}}$

EXPERIMENT	RESULT	ESTIMATE	COMMENT
a drawing pin dropped 100 times	point up 63	0.63	no counting could have found it
a die thrown 60 times	12 sixes	0.2	theory says $\frac{1}{6} \approx 0.167$
a coin tossed 200 times	92 heads	0.46	close to 0.5 — it looks fair

MORE TRIALS, BETTER ESTIMATE

Ten throws of a fair die can easily give four sixes; six hundred throws will not. A relative frequency far from the theory is evidence of bias only when the number of trials is large.

Once a probability is known, the expected number in n trials is n times it: 300 throws of a fair die should give about $300 \cdot \frac{1}{6} = 50$ sixes.

02

Worked examples

EXAMPLE 1 A bag holds 4 red, 3 blue, 2 green and 1 yellow counter. Find $P(\text{red})$ and $P(\text{not red})$.

1

There are 10 counters altogether.

2

$P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.

3

$P(\text{not red}) = 1 - \frac{2}{5} = \frac{3}{5}$.

Check: 6 of the 10 are not red ✓

ANSWER

$\frac{2}{5}$ and $\frac{3}{5}$

EXAMPLE 2 One fair die is thrown. Find $P(\text{even})$, $P(\text{more than 4})$ and $P(7)$.

① Even: 2, 4, 6 — three of six, so $\frac{1}{2}$.

② More than 4: 5, 6 — two of six, so $\frac{1}{3}$.

③ A 7 is impossible, so 0.
All three lie between 0 and 1 ✓

ANSWER

$\frac{1}{2}$, $\frac{1}{3}$ and 0

EXAMPLE 3 A drawing pin lands point up 63 times in 100 drops. Estimate the probability and the number of point-up landings in 500 drops.

① Relative frequency $\frac{63}{100} = 0.63$.

② Expected number = $500 \cdot 0.63$.

③ = 315.
An estimate, not a guarantee.

ANSWER

0.63 and about 315

03 Interactive model

Throw a die twenty times as a class and write the running relative frequency of sixes on the board; watching it settle towards a sixth is worth more than any explanation.

Numbers between 0 and 1

An impossible event has probability:

$P(4)$ on a fair die is:

$P(\text{even})$ on a fair die is:

$P(\text{more than } 4)$ on a fair die is:

$$\frac{1}{6}$$

$$\frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{2}{3}$$

With 4 red in 10 counters, $P(\text{not red})$ is:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{1}{2}$$

$$\frac{6}{4}$$

All the probabilities of one experiment add to:

$$0$$

$$\frac{1}{2}$$

$$1$$

$$100$$

12 sixes in 60 throws gives a relative frequency of:

0.12

0.2

0.6

5

In 300 throws of a fair die the expected number of sixes is:

6

30

50

60

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Event	Hodisa	Событие
Outcome	Natija	Исход
Equally likely	Teng ehtimolli	Равновозможные
Impossible	Imkonsiz	Невозможное
Certain	Muqarrar	Достоверное
Relative frequency	Nisbiy chastota	Относительная частота
Fair	Adolatli (bir jinsli)	Правильный (симметричный)

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Every probability lies:

above 1

between 0 and 1

below 0

anywhere

A certain event has probability:

0

$\frac{1}{2}$

1

100

The fraction rule needs the outcomes to be:

many

equally likely

few

written down

$P(\text{not } A)$ equals:

$$1 - P(A)$$

$$P(A) - 1$$

$$1 \div P(A)$$

$$0$$

Relative frequency is:

a theory

an estimate from experiment

always exact

a percentage

A better estimate comes from:

fewer trials

more trials

a bigger die

rounding

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The probability of an impossible event

ANSWER 0

2. The probability of a certain event

ANSWER 1

3. $P(\text{head})$ on a fair coin

ANSWER $\frac{1}{2}$

4. $P(4)$ on a fair die

ANSWER $\frac{1}{6}$

5. $P(\text{even})$ on a fair die

ANSWER $\frac{1}{2}$

6. $P(7)$ on a fair die

ANSWER 0

7. Every probability lies between

ANSWER 0 and 1

Medium · the standard question

7 PROBLEMS

1. $P(\text{more than } 4)$ on a fair die

ANSWER $\frac{1}{3}$

2. $P(\text{red})$ with 4 red in 10

ANSWER $\frac{2}{5}$

3. $P(\text{blue})$ with 3 blue in 10

ANSWER $\frac{3}{10}$

4. $P(\text{not red})$ for that bag

ANSWER $\frac{3}{5}$

5. The four probabilities of that bag add to

ANSWER 1

6. A pin lands point up 63 times in 100

ANSWER 0.63

7. 12 sixes in 60 throws

ANSWER 0.2

Hard · stretch and reason

7 PROBLEMS

1. $P(\text{a multiple of } 3)$ on a fair die

ANSWER $\frac{1}{3}$

2. $P(\text{not a } 6)$ on a fair die

ANSWER $\frac{5}{6}$

3. 5 red and 15 blue: $P(\text{red})$ as a percentage

ANSWER 25%

4. Is a die giving 12 sixes in 60 throws unfair?

ANSWER Probably not — 10 expected, 12 seen

5. Expected sixes in 300 throws of a fair die

ANSWER 50

6. $P(\text{red}) = 0.4$ in a bag of 25: the number of red

ANSWER 10

7. If $P(\text{win}) = 0.35$, then $P(\text{not win})$

ANSWER 0.65

07 Homework

Homework — 5 tasks

Every answer as a fraction in lowest terms or a decimal, and never outside 0 to 1.

1. A fair die is thrown. Find $P(3)$, $P(\text{odd})$ and $P(\text{at least } 5)$.
2. A bag holds 6 red, 9 blue and 5 green counters. Find $P(\text{red})$, $P(\text{blue})$ and $P(\text{not green})$.
3. Show that the three probabilities in task 2 add to 1.
4. A coin lands heads 92 times in 200 tosses. Estimate $P(\text{head})$ and say whether the coin looks fair.

5. If $P(\text{rain}) = 0.3$, find $P(\text{no rain})$ and the expected number of rainy days in 30.